BC and Federated Learning for the Management and Orchestration of Network Slices in 5G, Fog, Edge, and Clouds

Richa Golash, Shahnawaz Ahmad, Bhawana, and Naween Kumar

Abstract

The demand for secure, scalable, and low-latency data management solutions has intensified the integration of 5G networks, Edge computing, and Internet of Things (IoT)-driven technologies. Modern systems, characterized by distributed devices and sensitive data, face privacy, interoperability, and resource optimization challenges. Traditional centralized approaches struggle to address these issues due to vulnerabilities in security, high communication overhead, and regulatory constraints. This chapter explores integrating BC and Federated Learning (FL) technologies as a transformative paradigm to manage network slices in 5G, Fog, Edge, and Cloud environments, focusing on healthcare applications. Blockchain (BC) offers decentralized, tamper-proof data storage and Transaction (Tx) validation through consensus mechanisms like Proof of Work (PoW) and Proof of Stake (PoS). At the same time, FL enables collaborative Machine Learning (ML) without centralized data aggregation. Together, they form FedBLOC—a hybrid architecture that enhances trust, security, and efficiency in distributed systems. The paper systematically examines BC fundamentals, FL algorithms, and network slicing in 5G, highlighting their synergistic potential. Network slicing, which creates virtualized, application-specific networks, benefits from BC's immutability and FL's privacy-preserving model training, enabling real-time healthcare services like remote patient monitoring and Artificial Intelligence (AI)-driven diagnostics. By addressing challenges such as data security, incentive mechanisms, and system resilience, the integration of these technologies paves the way for robust, decentralized healthcare ecosystems. The chapter also discusses practical implementations, including secure medical record sharing, federated AI models for disease detection, and dynamic resource allocation in smart hospitals. This work underscores the critical role of BC and FL in advancing secure, efficient, and scalable solutions for next-generation networks and healthcare systems.

Keyword

BC • Federated learning • 5G • Network slicing • Healthcare • Edge computing • Cloud computing

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1 Introduction to Blockchain Technology

1.1 Basics and Applications in Healthcare

This section elaborates on Blockchain (BC) technology, its type, components, and applications in the healthcare domain are described as follows (Bhutta et al. 2021; Sarmah 2018).

1.1.1 Blockchain Basics

In 2008, an unknown person named Satoshi Nakamoto introduced a peer-to-peer electronic cash system in his paper that allows online payments without involving intermediaries (i.e., a centralized financial system) (Morkunas et al. 2019). The paper also stated the concept of digital currency that operates on a decentralized network, a public ledger to hold the digital currency Transaction (Tx) history, and cryptography to secure the Txs. Later, in 2009, he implemented the concept and proposed a Bitcoin cryptocurrency based on the distributed and decentralized system called BC. The BC is a growing list of continuous records assembled in blocks and linked together through a cryptographic hash called a chain of continuous blocks.

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N. Kumar et al. (eds.), *Integrating Cloud, Fog, and Edge Computing in Healthcare: Federated Learning and Blockchain Approaches*, Advances in Science, Technology & Innovation, https://doi.org/10.1007/978-3-031-96265-3_4



Fig. 1 Working of Bitcoin BC

Figure 1 illustrates the working of a Bitcoin BC between two actors (i.e., Alice and Bob) for transferring cryptocurrency, as follows:

(a) Initiate Tx: Alice initiates a Tx converted into a hashed Tx using cryptographic hash functions such as Secure Hash Algorithm (SHA256), keccak256, SHA512, etc. The proposed Tx contains the timestamp, date, quantity, sender, and receiver addresses.
(b) Add Cryptographic Digital Signature: The proposed Tx is stamped with a unique cryptographic digital signature, such as an Elliptic Curve Digital Signature Algorithm, to ensure the authenticity and integrity of the Tx.
(c) Authenticate and Append Proposed Tx: The proposed Txs are collected by miners to solve the cryptographic puzzle called mining and generate a valid block, then broadcast to the decentralized network for the authentication process. The block contains the previous, hash, Merkle root hash, nonce, block hash, timestamp, and miner address.
(d) Broadcast Tx to Distributed Network: The computers process the Txs, verify them, and once authenticated, the Tx is included in the distributed ledger/BC through the block.
(e) Finalize Tx: Bob receives the Bitcoin cryptocurrency once the proposed Tx is added to the BC. These proposed Txs are linked with the previously added records to ensure immutability. Also, provides a complete, variable, and irreversible history of Txs made on the BC.

After the evolution of the Bitcoin BC, in 2013, Vitalik Buterin proposed the Ethereum 1.0 BC operated on Proof of Work (PoW) with Ether cryptocurrency in his whitepaper. It went live in 2015 to design and implement various BC-based applications/use cases using smart contracts logic, which was impossible with the Bitcoin BC. It also supports distributed storage through the Interplanetary File System (IPFS), Swarm, and Filecoin to store files, images, audio, video, etc., and their link stores on the Ethereum BC in the form of hash. Later, in 2022, Ethereum 2.0 came up with the Proof of Stake (PoS) consensus algorithm to improve the scalability, speed, security, and efficiency of the Ethereum 1.0 BC (Guo and Yu 2022). In 2015, the Linux Foundation announced an open-source software for building enterprise BC solutions called Hyperledger Project. It consists of various frameworks such as Hyperledger Indy, Hyperledger Aries, Hyperledger Besu, Hyperledger Iroha, etc. After that, various BCs were designed and developed to give enterprise solutions, such as Ripple (2016, integrating BC with traditional banking system), EOS. IO (2017, focus on scalability and user-friendliness), Polkadot (2020, enable interoperability), Avalanche (2020, mining high throughput and low latency), Solana (2020, address scalability issue), etc. (Sanka et al. 2021).

1.1.2 Blockchain Types

The BC is broadly divided into two categories, i.e., permissionless and permissioned, as shown in Fig. 2.

- Public BC: Public blockchains are permissionless. Therefore, anyone can join the BC network to perform read, write, and consensus operations without prior permission. It is fully distributed and decentralized due to this selfish mining, 51% attack, double spending, money laundering, and tracking are possible, e.g., Bitcoin BC, and Ethereum BC.
- Private BC: The private BC is permissioned and hence requires prior permission to join the BC network. The authorized person can perform read, write, and validation operations. It is intended for enterprises to handle scalability, privacy (such as channels and digital identities), and security (including 51%, selfish mining, flooding, and distributed denial of service attacks) issues, e.g., Hyperledger, Multichain, Blockstack, and Ripple.
- Consortium/Federated BC: The consortium BC is permissioned, that is, controlled and managed by a group of authorized persons. Only pre-selected authorized persons can perform consensus operations, whereas the rest

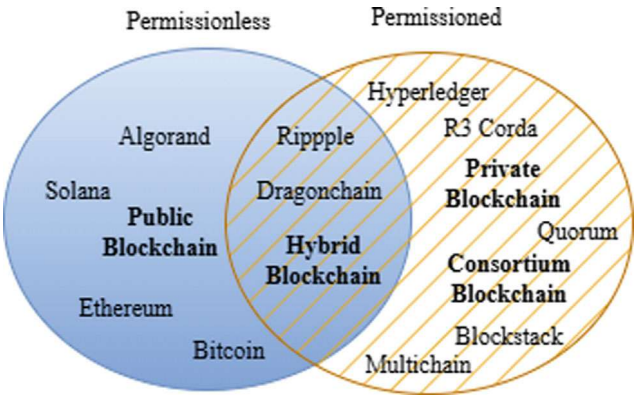


Fig. 2 BC classification

can carry out read and write operations. It is partially centralized as compared to the private BC, e.g., Hyperledger, R3 Corda, and Quorum.
- Hybrid BC: The hybrid BC contains the feature of permissionless and permissioned, which allows some parts of the BC to be accessible for the public to perform read and write operations. While the rest is restricted, e.g., Ripple.

1.1.3 Components of Blockchain

- Smart Contract: The smart contract, also known as chaincode, is a self-executable program that consists of rules and decision stored on the BC. In 1994, Nick Szabo, a computer scientist, introduced the concept of a smart contract as "*computerized Tx protocols that execute the terms of a smart contract*". In 2015, smart contracts became a reality because of the Ethereum BC that allows the writing of complex programming scripts using Solidity language that respond to certain types of events. The goal of the Ethereum BC was to break the limitations of the Bitcoin BC using smart contracts.
- Block: The block stores data of the BC in a structural format that contains a block header and block body. The block header comprises the block version, Merkle tree root hash, nBits (difficulty target), nonce, timestamp, and previous block hash are explained as follows:

  (a) Block Version: This field specifies the set of rules (i.e., BC protocol's version: BIP) to follow for validating a block.
  (b) Merkle Tree Root Hash: This field contains the hash of all Txs in a block for efficient verification of Txs.
  (c) nBits: This field specifies the difficulty level for a valid block to ensure blocks are added at a consistent time interval.
  (d) Nonce: The nonce is a 4-byte field that contains a random value adjusted by miners to identify a valid block hash that meets the difficulty target.
  (e) Timestamp: This field records the exact time (i.e., universal time) when the block was created to maintain the chronological order.
  (f) Previous Block Hash: This file contains a 256-bit hash value that points to the previous block header, ensuring immutability.

  The block body contains the set of Txs and Tx counters. The Tx further includes Tx ID (unique identification number for each Tx), digital signature, sender and receiver's information (public addresses), and smart contract or chaincode data (i.e., optional depending on the type of BC used). The maximum number of Txs a block can hold depends upon the size and size of each Tx or block size.
- Consensus Protocol: The consensus protocol is a set of mutual agreements (rules) to synchronize the BC among BC nodes (i.e., participants). It also explains how the new blocks are stored on the BC, where each BC node does not trust or know each other (Zheng et al. 2017).

  (a) Byzantine Fault Tolerant (BFT): The BFT consensus emerges from the traditional Byzantine Generals Problem (BGP), which was originally presented in 1982. The problem describes how the group of generals come to a common battle plan in the presence of traitors for misleading loyal generals. To come up with a solution to this problem, the BFT consensus was proposed to handle faulty nodes and work on a Byzantine agreement to reach a common agreement. However, this algorithm can only tolerate 1/3 of faulty nodes among the total participating nodes.
  (b) Practical Byzantine Fault Tolerance (PBFT): The PBFT is based on classical BFT state machine replication and voting-based consensus protocol, which considers safety and liveliness to guarantee the correctness and tolerance of Byzantine faults. In each round, a leader node is selected based on some predefined rule for ordering the Txs. This process is divided into three phases, i.e., pre-prepared, prepared, and committed. In each round, a leader node requires 2/3 votes of the total nodes present in the network to enter into the next phase. PBFT does not consider any hashing procedure and tolerates up to 1/3 of malicious nodes among a total number of nodes present in the BC environment to achieve consensus. A similar concept is presented in Delegate Byzantine Fault Tolerance (dBFT), where only some delegated nodes are voted to record the Tx instead of all.
  (c) Proof of Work (PoW): In PoW, miners have to solve a cryptographic puzzle, also known as a difficulty target. The difficulty target is a hash function that contains some leading zeros. The miner selects a set of Txs from the mempool where unconfirmed Txs reside, nonce, and timestamp to generate the hash value that should be less than or equal to the difficulty target. The miner solves the cryptographic puzzle once the hash value is generated based on the given criteria. He/she replicates the block to the other miners for verification. Once the verification is done, the block is valid and adopted by other miners in the BC.
  (d) Proof of Stake (PoS): The PoS is an energy-efficient alternative of PoW in which validators use his/her stake power to generate new blocks in the BC and receive Tx fees as incentives. Otherwise, for an invalid block, the validator loses his/her stakes. The validators are shortlisted either using coin age selection, randomized selection, or weighted random selection (i.e., stake weighting).
  (e) Delegate Proof of Stake (DPoS): The DPoS validators elect their delegate instead of depending on the wealthiest condition to generate the block. However,

only a few nodes validate the block so that the block confirmation time is reduced and Txs are confirmed quickly. Meanwhile, the block size and block interval are tuned automatically. Using this consensus process, participating nodes need not worry about dishonest delegate nodes, and if something malicious is found, then the delegate is voted out easily.

(f) Proof of Capacity (PoC): The PoC is based on the storage space concept instead of buying expensive computational hardware as in PoW to make it energy efficient. The eco-efficient and egalitarian features of PoC make it different from the other consensus protocols in solving the biggest problems faced by other cryptocurrency platforms, such as scalability and the lack of storing legacy blocks on the BC without overloading the storage capacity. It uses the available storage space on the hard disk by configuring it in a process called plotting. The more plots on the hard disk will give the chance to solve the next block in the BC.

(g) Proof of Burn (PoB): The PoB consensus protocol is based on the concept that the participant has to burn the coin to mine the block in the BC. The miner has to send their coin to the eater's address so that it cannot be accessed or spent again and get rewarded as a result of this action.

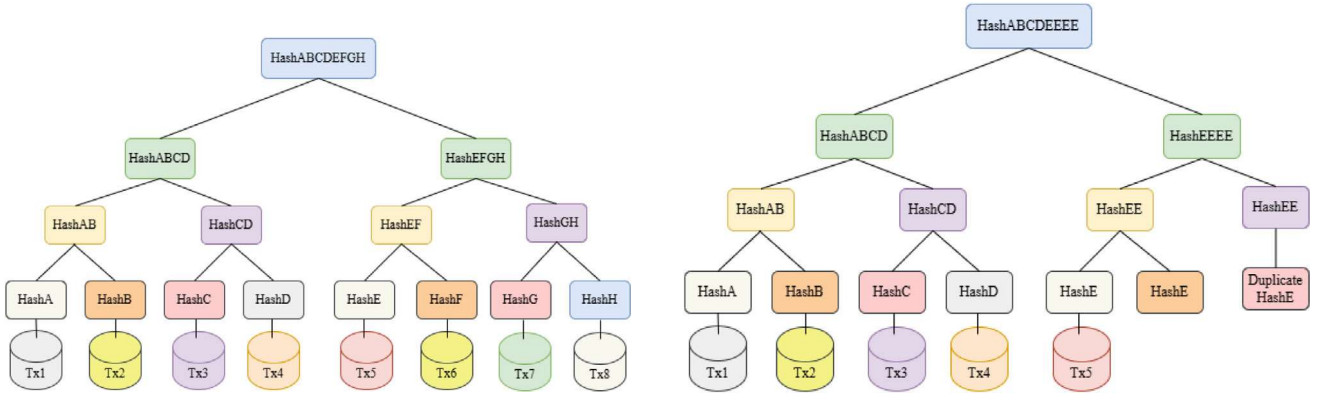
(h) RAFT: The RAFT consensus protocol is derived from PAXOS and based on election process, to achieve agreement nodes collectively elect a leader in the network and rest of the network behaves as followers sometime called as replicated state machine. In each round, a candidate node sends the proposal to become a leader. In the next round, it has to go through three processes: (i) voting request, (ii) electing the leader, and (iii) elected leader. Once the leader is elected, then it will propose Txs in the network to the follower based on a majority for acceptance, and the proposed Tx is committed.

(i) Tendermint: The tendermint is based on Byzantine consensus. The proposer selection process is done to broadcast the unconfirmed block in each round. All nodes need to know about the proposer selection process, which includes three steps these are pre-vote, pre-commit, and commit step. In pre-vote, the client will disseminate their Txs to validator nodes using the gossip protocol. The validator decides whether to add a Tx to the block by broadcasting the pre-vote for the proposed block. If the node gets more than 2/3 of the pre-vote for the proposed block, then it enters into the commit phase. In the commit phase again the node will validate the block and broadcast the commit for that block, if node receive 2/3 commit of the proposed block then the block will commit otherwise discarded.

- Merkle Tree: The Merkle tree, also known as a hash tree, provides efficient and secure Tx verification by arranging the Tx in the form of a tree. In this tree structure, the leaf node represents the Tx hash, and the non-leaf node indicates the combined hash value of the child node (i.e., leaf or non-leaf). The Merkle tree is categorized into the balanced and unbalanced Merkle tree, as shown in Fig. 3a, b. In a balanced Merkle tree, two different hash values are combined to form a root hash, whereas in an unbalanced Merkle tree, the same hash value is replicated to form a binary structure to generate a root hash.
- Digital Signature: The digital signature is utilized to authenticate Txs and ensure Proof of Authority (PoA). It is applied either using the Edwards Curve digital signature algorithm, Elliptic Curve cryptography, or Elliptic Curve digital signature algorithm. It typically includes two phases, i.e., the signing phase and the verification phase. In the signing phase, the signer uses its private key to encrypt the data. Whereas, in the verification phase, the recipient verifies the signature and data using the signer's public key and decrypts the original information.

1.1.4 Blockchain Applications in Heathcare

In Capece and Lorenzi (2020), authors proposed a permissioned BC model based on PoA consensus to store and manage Electronic Healthcare Records (EHR) of patients in Sistema Sanitario Nazionale (SSN), the Italian National Health Service. Using this model, registered patients can easily access their personal medical data through their smartphones anywhere at any time. In Azaria et al. (2016), authors presented a decentralized BC record management system called MedRec to manage electronic health records. The MedRec is designed and implemented using the Ethereum BC that uses the PoW consensus algorithm to incentivize medical stakeholders to add new blocks. Also, it contains a register, patient-provider relationship, and summary contracts to perform various operations on the MedRec. In Zhou et al. (2018), authors designed and proposed a BC-enabled medical insurance storage system using the Ethereum BC called MIStore. The proposed system performs patient spending data verification using homomorphism computation. In Du et al. (2020), authors proposed a consortium BC-based medical record-sharing framework to prevent Tx manipulation and fraud. They designed a new consensus algorithm known as the mixed BFT algorithm and supervised an anonymous information-sharing process. So that only relevant nodes can view the Tx details. In Liu et al. (2019), authors proposed a private BC-enabled medical data sharing and protection scheme to improve the EHR system for hospitals. They designed an improved DPoS consensus algorithm and symptom matching for patients. So that they can communicate in



(a) Balanced Merkle Tree (b) Unbalanced Merkle Tree

Fig. 3 Types of Merkle tree

the future about the disease information based on a shared session key. In Daraghmi et al. (2019), authors proposed an Ethereum BC-based system called MedChain for managing medical records. Additionally, they designed a PoA-enabled incentive mechanism to create new blocks and timed-based smart contracts for controlling access to Electronic Medical Records (EMR). In Hang et al. (2019), authors designed a permissioned BC-enabled medical DApp to secure the EMR using smart contracts. Further, they designed access control policies for authorized participants to access Tx records. In Madine et al. (2020), the authors proposed an Ethereum BC-based smart contracts system that provides control to patients over their medical records. The proposed system integrates an interplanetary file system and reputation-enabled re-encryption oracles to securely fetch, store, and share patients' medical records. In Hovorushchenko et al. (2022), the authors developed a medical data management based on the BC for accessing the sufficiency of medical data. It consists of four groups of users, i.e., users (patients that upload medical data to the system), validators (check the quality and authenticity of uploaded data), clients (doctors that compile and study patient reports), pharmaceuticals (research companies that access user data) through BC. In Kassou et al. (2021), the authors proposed a BC-based architecture for medical and water waste management. For tracking water and waste management, various Internet of Things (IoT) devices are equipped that monitor and communicate information on the BC through consensus (i.e., DPoS) about water and containers of waste after treatment.

2 Federated Learning Overview

This section presents the overview of Federated Learning (FL) and its classification, along with its algorithm and use in the healthcare sector, is explained as follows:

2.1 Definition and Mechanisms

FL is a decentralized Machine Learning (ML) approach that enables multiple devices or nodes to collaboratively train a shared model without exchanging raw data. Instead of centralizing data in a single repository, FL allows data to remain on local devices, ensuring privacy and reducing communication overhead. A central server aggregates model updates from participating clients, enhancing scalability and security. This paradigm is particularly useful in distributed environments such as 5G, Fog, Edge, and Cloud Computing, where data is generated at multiple locations. By leveraging FL, networked systems can optimize resource allocation, improve security, and enable intelligent decision-making without compromising user privacy.

Devices using FL jointly modify models under decentralized and centralized control but preserve data confidentiality at the individual level. Devices under FL maintain local storage of their data through distributed data spreading rather than moving data to centralized databases. A participant device relies on its retained dataset during training operations that prevent them from disclosing their actual information. Device-connected FL mechanisms enable centralized server reception of model updates containing optimized parameters and gradient adjustments for collaborative overall model construction while ensuring user data confidentiality. The approach protects data privacy during specific model customization processes because it functions efficiently with privacy-sensitive operations. The training process takes place directly on devices to keep data secure as model updates combine into a unified global model that outperforms traditional data center approaches. The technology enables devices to learn by receiving model updates rather than exchanging data to preserve privacy (AbdulRahman et al. 2020; Khan et al. 2021). FL emerged through the development of decentralized computing technology and device-based edge learning

to represent a new fundamental learning design. By maintaining training data locally at edge servers and devices, this method creates safeguards to protect sensitive information. The FL system establishes client device privacy safeguards while resolving ongoing storage issues and communication complications. Some of the key benefits of FL include

(a) Offline functionality
(b) Improved latency performance
(c) Higher accuracy
(d) Strengthened privacy measures
(e) Extended device battery life
(f) Localized model training.

The performance data combined with technical challenges related to Edge computing AI/ML approaches can be found in Table 1 together with assessments about distributed learning capabilities. The successful executions of TOFEL and DML reached more than 90% detection accuracy during diverse dataset evaluations requiring compression method optimization. Research must focus on meticulous parameter optimization and developing protective control methods before the FEDGS system can securely implement Semi-supervised learning. Upcoming studies will evaluate multiple Edge devices using FFT alongside efforts to enhance vehicle ecosystems integrated with DQN, as well as studies on improving convergence between FL built from deep Reinforcement Learning (RL) and self-supervised learning. To establish robust security measures and effective deployment of FL across different Edge locations, new technological improvements must be implemented.

2.1.1 Types of Federated Learning Algorithms

FL is classified into five categories, as shown in Fig. 4 and are described below:

(a) Weighted Federated Learning: The weighted federated learning algorithm optimizes global model training by incorporating domain adaptation through feature discrepancy minimization, as shown in Algorithm 1. The server initializes the global model and orchestrates multiple training rounds, where each participating client updates its local model using its dataset. Clients extract source features from their local data and generate target features using a separate target dataset. The MMD between source and target features is computed to determine aggregation

Table 1 Comparison of AI/ML approaches and future challenges in FL

Reference	Accuracy >90%	AI/ML approach	Open areas and future challenges
Huang et al. (2022)	YES	TOFEL	Evaluating FL-PQSU with fresh datasets and IoT applications, exploring alternative compression mechanisms
Li et al. (2022)	NO	FEDGS	Investigating hyperparameter optimization for efficient branch designs for Big.Little architecture
Yang et al. (2021)	YES	DML	Improving E-Tree efficiency by refining layer depth and aggregation frequency using reinforcement learning
Math et al. (2021)	NO	Semi-supervised	Developing robust control strategies for fuzzy logic-based cooperative game theory and advanced control methods
Orescanin et al. (2021)	NO	FFT	Experimenting with alternative edge devices and integrating additional sensor-equipped clients for improved networking
Li et al. (2021)	–	DQN	Enhancing vehicular edge networks with improved data migration and fault-tolerant public infrastructure
Sun et al. (2022)	–	FL	Exploring self-supervised learning and deep RL for improved convergence in complex domains

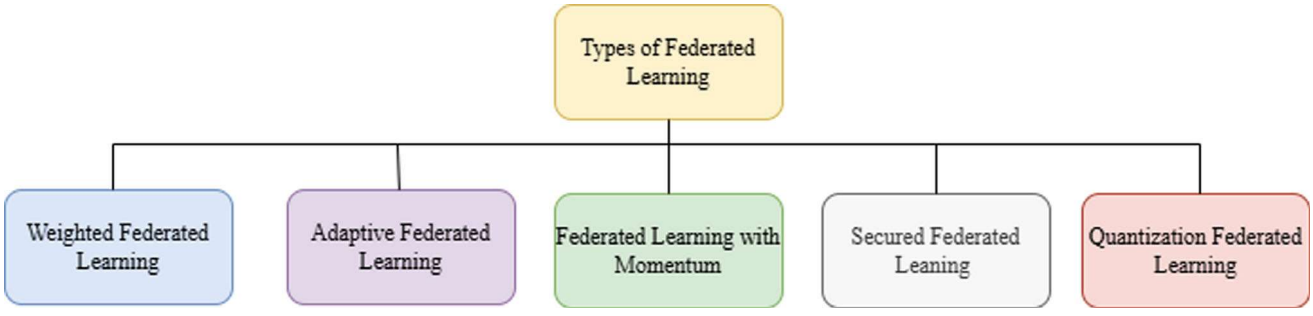


Fig. 4 Classification of federated learning algorithms

weights, ensuring that clients with lower feature discrepancies contribute more to the global model. The global model is then updated as a weighted sum of local models. On the client side, local training is performed through multiple epochs, updating model parameters using gradient descent. The final global model is obtained by aggregating the weighted contributions from all clients, improving generalization across distributed data sources (Chen et al. 2022).

Algorithm 1: Weighted Federated Learning

Server-Side Execution

Initialize global model parameters: $\omega_{G,0}$

for *each global epoch* $t = 1$ *to* E_{global} **do**

  for *each participating client* k *in parallel* **do**

    Compute updated local model $\omega_{k,t}$ and source features $f_{S_k,t}$ via

    Client_Local_Update($\omega_{G,t-1}, k$)

    Obtain target features $f_{T_k,t}$ via

    Target_Client_Feature($\omega_{k,t}, X_T$)

    Compute discrepancy $MMD_{k,t}$ between source and target features:

$$MMD_{k,t} = \left\| f_{S_k,t} - f_{T_k,t} \right\|^2$$

    Compute aggregation weight:

$$\alpha_{k,t} = \frac{1/MMD_{k,t}}{\sum_{k=1}^{N} 1/MMD_{k,t}}$$

    Aggregate global model:

$$\omega_{G,t} = \sum_{k=1}^{N} \alpha_{k,t} \omega_{k,t}$$

Client-Side Execution

Function `Client_Local_Update(`$\omega_{G,t-1}, k$`)`**:**

  Initialize client model $\omega_{k,t-1} = \omega_{G,t-1}$

  for *each local epoch* $i = 1$ *to* E_{local} **do**

    for *each batch in local dataset* **do**

      Update model parameters:

$$\omega_{k,t} = \omega_{k,t-1} - \eta \nabla_{\omega_{k,t-1}} L(b_{k,t-1})$$

      Extract source features:

$$f_{S_k,t} = F_{\omega_{k,t}}(X_{k,S})$$

  return $\omega_{k,t}$ and $f_{S_k,t}$ to server

Function `Target_Client_Feature(`$\omega_{k,t}, X_T$`)`**:**

  Generate target features:

$$f_{T_k,t} = F_{\omega_{k,t}}(X_T)$$

  return $f_{T_k,t}$ to server

Result: Global Model ω_{global}

(b) Adaptive Federated Learning: The adaptive federated learning procedure dynamically adjusts learning rates to enhance convergence while preserving privacy, as shown in Algorithm 2. It begins by initializing parameters and iteratively computing model updates through gradient estimation. At each step, it calculates first and second moment estimates to refine learning rate adjustments using a clipping function. These refined estimates help stabilize updates, ensuring smooth convergence. The local models continue training until convergence criteria are met, after which they send updates to a central aggregator. The aggregator computes and distributes contributions across clients, enabling collaborative learning. The process repeats until an optimized global model is obtained (Wu et al. 2022; Wang et al. 2019).

Algorithm 2: Adaptive Federated Learning

Input: Training dataset, privacy constraint ϵ, step size η

Output: Model parameters Θ

Initialize: Set $\lambda_{1,0} = 0$, $\lambda_{2,0} = 0$

for *each iteration round* **do**

  Perform gradient computation at step t: $\gamma_t \leftarrow \nabla_\Theta g_t(\Theta_{t-1})$

  Compute first-moment estimate:

$$\lambda_{1,t} \leftarrow \theta_{t1}\lambda_{1,t-1} + (1 - \theta_{t1})\gamma_t$$

  Compute second-moment estimate:

$$\lambda_{2,t} \leftarrow \theta_2 \lambda_{2,t-1} + (1 - \theta_2)\gamma_t^2$$

  Adjust learning rates using clipping function:

$$\text{Clip}\left(\frac{\alpha}{\sqrt{\lambda_{2,t}}}, \eta_l, \eta_u\right)$$

  Refine first-moment estimation:

$$\bar{\lambda}_{1,t} = \frac{\lambda_{1,t}}{1 - \theta_{t1}}$$

  Refine second-moment estimation:

$$\bar{\lambda}_{2,t} = \frac{\lambda_{2,t}}{1 - \theta_2}$$

  Until local dataset achieves convergence

  Send model updates Θ_t of the current round to the central aggregator

  The central aggregator computes the contribution:

$$\Theta_i - \Theta_{i-1}$$

  and distributes it accordingly

Return final parameters $\Theta_{i1}, \Theta_{i2}, ..., \Theta_{iK}$ from the clients to the central server

(c) Federated Learning with Momentum: The federated learning with momentum algorithm enhances traditional FL by incorporating a momentum-based optimization

strategy to improve convergence and stability, as shown in Algorithm 3. It begins by initializing the global model, communication rounds, and momentum parameters. In each communication round, a subset of devices is randomly selected to participate in local training. Each device updates the model iteratively using mini-batches over multiple local epochs. After local training, weighted model updates from all participating devices are aggregated to compute a global update. A momentum term is introduced, which combines past updates with the current one, smoothing the learning process and reducing oscillations. Finally, the global model is updated using the momentum-adjusted update, improving training efficiency and convergence in decentralized learning environments (Salazar et al. 2023).

Algorithm 3: Momentum Federated Learning

Input: Global model Θ_k, number of communication rounds k, momentum parameter
Output: Updated global model Θ
Initialize: Set momentum coefficient μ_k (with $\mu_0 = 0$), learning rate λ, mini-batch size M, and local training epochs T.

for *each communication round k* **do**
  Randomly select a subset of devices $\mathbb{D}_k$ from the total set of devices $\mathbb{D}$

  for *each participating device* $d \in \mathbb{D}_k$ **do**
    for *each local epoch* $j = 1$ *to* T **do**
      for *each mini-batch* $m \in M$ **do**
        Update model:

$$\Theta_{k+1,x} = \Theta_k - \lambda \nabla_x(\Theta_k, m)$$

  Compute the weighted update:

$$\Delta = \sum_{x \in \mathbb{D}_k} \left(\frac{n_x}{N} (\Theta_{k+1,x} - \Theta_k) \right)$$

  Update momentum:

$$\mu_{k+1} = \tau \mu_k + (1 - \tau)\Delta$$

  Apply momentum to the global model:

$$\Theta_{k+1} = \Theta_k + \mu_{k+1}$$

Result: Updated global model Θ

(d) Secure Federated Learning: The secure federated learning algorithm ensures privacy-preserving model training by incorporating periodic noise addition and secure aggregation, as shown in Algorithm 4. Initially, the central Parameter Server (PS) broadcasts the global model to all clients. In each iteration, selected clients update their local models by computing gradients on mini-batches of data. If the iteration is a multiple of a predefined interval Q, clients synchronize their models with the global model; otherwise, they perform local updates. When an iteration reaches the synchronization point, clients add Gaussian noise to their updated models before sending them to the PS, enhancing privacy. The PS then aggregates the received models and broadcasts the new global model to the next subset of selected clients. This process continues iteratively, ensuring secure and efficient federated learning while preserving data privacy (Li et al. 2020).

Algorithm 4: Secure Federated Learning

Input: Initial model $\bar{w}_0$, Step size η_0
Output: Optimized global model $\bar{w}_T$
PS broadcasts $\bar{w}_0$ to all clients ($S_0 = [N]$)

for *each iteration* $t = 0, 1, ..., T-1$ **do**
  Client-side:
  for *each client k in the subset* S_t **do**
    if *mod*$(t, Q) = 0$ **then**
      Set client's model $w_k^t = \bar{w}^t$
    else
      Sample mini-batch ξ_k^t, compute local gradient:

$$g_k^t = \nabla F_k(w_k^t; \xi_k^t, b)$$

    if *mod*$(t+1, Q) \neq 0$ **then**
      Update model:

$$w_k^{t+1} = w_k^t - \eta^t g_k^t$$

    else
      Add noise:

$$w_k^{t+1} = (w_k^t - \eta^t g_k^t) + z_k^t, \quad z_k^t \sim \mathcal{N}(0, \sigma_k^{2t} I_m)$$

      Send w_k^{t+1} to PS
  for *clients not in* S_t **do**
    Update model: $w_k^{t+1} = w_k^t$
  Server-side:
  if *mod*$(t+1, Q) = 0$ **then**
    Aggregate models:

$$\bar{w}^{t+1} = \frac{1}{K} \sum_{k \in S_t} p_k w_k^{t+1}$$

    Sample subset S^{t+1}, broadcast $\bar{w}^{t+1}$ to all clients

Result: Final optimized global model $\bar{w}_T$

(e) Quantization Federated Learning: The quantization federated learning algorithm enhances communication efficiency by reducing model update sizes through quantization at both client and Edge levels, as shown in Algorithm 5. Initially, a global model is initialized on the Cloud server and distributed to Edge servers. Clients perform local model updates iteratively using gradient descent while keeping data decentralized. After completing local training, clients quantize and send their model updates to the respective Edge servers, which

aggregate them and further quantize updates before forwarding them to the Cloud server. The Cloud server then integrates these quantized updates to refine the global model. This hierarchical quantization approach reduces communication overhead while maintaining model accuracy, making it suitable for resource-constrained distributed learning environments (Liu et al. 2022).

Algorithm 5: Quantization Federated Learning

Input: Initial model on the cloud server X_0
Output: Final trained model X_K
for *each iteration* $k = 0, 1, ..., K-1$ **do**
  for *each edge server* $u = 1, ..., s$ ***(run in parallel)*** **do**
    Set the edge model $u_{u,k,0} = X_k$
    for *each sub-iteration* $t_2 = 0, 1, ..., \tau_2 - 1$ **do**
      for *each client* $i \in C_u$ ***(run in parallel)*** **do**
        Set the client model $x_{i,k,t_2,0} = u_{u,k,t_2}$
        for *each inner iteration* $t_1 = 0, 1, ..., \tau_1 - 1$ **do**
          Update the client model:

$$x_{i,k,t_2,t_1+1} = x_{i,k,t_2,t_1} - \eta \nabla f_i(x_{i,k,t_2,t_1})$$

        Send quantized update to the edge server:

$$Q_1(x_{i,k,t_2,\tau_1} - x_{i,k,t_2,0})$$

      Edge server aggregates:

$$u_{u,k,t_2+1} = u_{u,k,t_2} + \frac{1}{m_u} \sum_{i \in D_u} Q_1(x_{i,k,t_2,\tau_1} - x_{i,k,t_2,0})$$

    Send quantized update to the cloud server:

$$Q_2(u_{u,k,t_2} - u_{u,k,0})$$

  Cloud server aggregates:

$$X_{k+1} = X_k + \sum_{u=1}^{s} \frac{m_u}{n} Q_2(u_{u,k,t_2} - u_{u,k,0})$$

Result: Final trained global model X_K

2.2 Applications in Health Data

The fast development of smart healthcare technologies sustains early dementia detection through cost-effective and accessible methods. Healthcare professionals now face more pressure due to COVID-19 pandemic conditions while needing to provide remote diagnosis and treatment to their patients. Modern healthcare devices with IoT capabilities have gained extensive adoption because people focus more on their health. The growing reliance on technology through the Internet of Medical Things (IoMT) has exposed the system to cyber security risks.

A Secured Healthcare System (SHCS) must be established urgently because it provides essential protection for medical data against various cyber threats. Various measurable medical data flows securely into central databases through integrated sensor and wearable technology platforms operated by numerous developed nations. FL emerges as an effective privacy-preserving solution for the development of predictive health models across distributed settings. FL enables medical institutions to work with disbursed healthcare data sources while maintaining privacy standards for advancing disease management techniques in decentralized care settings.

In Shen et al. (2023), the authors developed an efficient privacy-protected diagnosis system for online e-healthcare using FL. The system resolves healthcare data privacy and security issues through distributed device-based model training without exposing raw patient data to other devices. Through their FL implementation the authors provide data protection together with diagnostic precision at high levels. The scheme implemented privacy-preserving methods that guard medical data secrecy throughout the learning process which makes it appropriate for e-healthcare applications in practice. The proposed methodology delivers better performance and security for protected online medical assessment through an efficient solution. Extensive experimentation has proven the validity of this work, which effectively balances e-healthcare system privacy requirements with performance standards. In Hossein et al. (2021), the authors proposed a novel BC-based privacy-preserving architecture called BCHealth for IoT healthcare applications to enhance privacy and security. BCHealth addresses the challenges of data integrity, confidentiality, and secure data sharing by leveraging BC to create a decentralized and tamper-proof architecture. The proposed solution ensures secure storage and transmission of sensitive medical data while maintaining patient privacy through cryptographic techniques and smart contracts. The framework is validated through performance evaluations, demonstrating its effectiveness in providing robust privacy-preserving mechanisms for IoT-enabled healthcare environments. In Alzubi et al. (2022), the authors introduced a privacy-protecting Cloud-IIoT-based EHRs framework that uses CNNs and BC-powered FL for combined model training without exposing complete patient data. The approach implements BC along with FL to preserve data security through decentralized trust management systems and model training procedures that do not require raw patient data exchange. The method delivers superior privacy protection together with reduced communication overhead and superior model performance than what traditional centralized learning techniques offer. The framework successfully achieves secure EHR analysis with integrity maintenance through experimental validation of its operational effectiveness. In Singh et al. (2022), the authors proposed a novel hierarchical FL framework for intrusion detection on the IoMT using a dew-Cloud architecture. By leveraging a multi-tiered approach, the framework efficiently processes data at the Edge (dew layer) before aggregating insights in the Cloud, reducing latency

and enhancing privacy. The proposed model integrates deep learning techniques to improve anomaly detection accuracy while minimizing communication overhead. Experimental results demonstrate significant improvements in detection performance and resource efficiency compared to traditional centralized and flat FL approaches. This work highlights the potential of hierarchical FL in securing IoMT networks by balancing security, scalability, and computational efficiency. In Kalapaaking et al. (2022), the authors presented a Secure Multi-Party Computation (SMPC)-based FL framework for 6G-enabled IoMT to enhance data privacy and security in decentralized healthcare applications. The study integrates SMPC with FL to ensure that model training occurs without exposing sensitive medical data, addressing key challenges such as privacy-preserving computation, communication efficiency, and model accuracy in distributed healthcare networks. Through experimental evaluations, the proposed approach demonstrates improved security and performance, making it a viable solution for privacy-aware AI-driven healthcare applications in 6G-enabled IoMT environments. In Kalapaaking et al. (2022), the authors established a CNN-based FL system that employs SMPC for encrypted inference together with encrypted aggregation processes. The proposed system enables several healthcare facilities and IoMT devices to integrate Edge computing clusters that encrypt local models before cloud-based SMPC-based aggregation. Subsequently, the encrypted global model is deployed to edge servers, which perform localized training procedures to boost accuracy levels. Hospitals maintain data and model privacy with encrypted inference capabilities through their choice of conducting operations on Cloud resources or Edge servers. Healthcare organizations benefit from a confidential model training and inference framework that enables distributed work across various network environments. The devices are attributed to hospital Edge or Cloud systems, which enable results to reach physicians located at multiple hospitals. In Fourati and Samiha (2021), authors emphasized FLs vital function in the IoMT system throughout the COVID-19 pandemic. They established principles of smart healthcare system architecture through Fog layer examination together with descriptions of pre-processing operations that involve ML and DL procedures at the Fog level. Further, performed an assessment of FL implementations across different COVID-19 application domains.

3 Network Slicing in 5G Environments

In 2015, the Next Generation Mobile Network (NGMN) alliance first introduced the concept of network slicing to address the heterogeneous service requirements for 5G and beyond (Li 2017). The objective of network slicing is to create multiple network slices (i.e., virtual network) on top of the shared physical network to host dedicated 5G applications. The network slicing architecture contains three layers, i.e., infrastructure layer, network layer, and service layer are described as follows:

(a) Infrastructure Layer: The infrastructure layer provides physical as well as virtual resources to 5G applications, including storage, computing, and network connectivity.
(b) Network layer: The network layer is responsible for performing operations related to configuration, Lifecycle Management (LCM), and deployment of network functions. In this layer, Software Defined Networking (SDN) is utilized for programmable network control, whereas Network Function Virtualization (NFV) is used for deployment of Virtual Network Functions (VNFs) or Cloud-Native Network Functions (CNFs).
(c) Service Layer: The service layer runs high-speed, low latency, and highly reliable 5G applications, including industry automation, connected vehicles, AR/VR, e-healthcare, etc.

Apart from this, the Management and Orchestration (MANO) is an important component in network slicing that is used to coordinate and manage the life cycle of the network slices. It also acts as a supervisor that ensures network functions and virtual resources are working properly to meet the requirements of different network slices (Kaloxylos 2018; Olimid and Nencioni 2020).

3.1 Different Types of Network Slicing Based on Functionality

Network slicing is broadly classified into two categories based on its functionality: service-based network slicing and deployment-based network slicing, as shown in algorithm Fig. 5. The service-based network slicing particularly focuses on performance characteristics such as speed, latency, and connectivity. On the other hand, deployment-based network slicing focuses on the structure and management of slices across the network (Khan et al. 2020). Both the categories are explained as follows.

3.1.1 Service Based Network Slicing

The service-based network slicing divides the 5G network into multiple virtual slices to run a specific 5G application. Each virtual slice is constructed to meet different Quality of Service (QoS) requirements, such as latency, bandwidth, speed, and reliability for each 5G application.

(a) Enhanced Mobile Broadband (eMBB): The eMBB is designed for high bandwidth and high-speed 5G applications that require low latency and fast data processing.

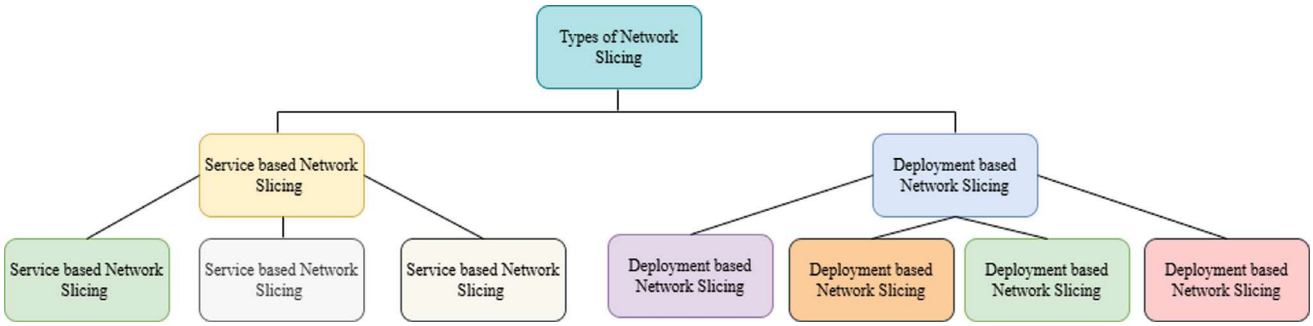


Fig. 5 Classification of network slicing

It is often used for HD video consultation, medical training, medical image transferring, and disease diagnostics.

(b) Ulta-Reliable Low-Latency Communication (URLLC): The URLLC is designed for mission-critical 5G applications that require real-time and ultra-reliable communication. It is often used for robot-assisted surgery, emergency patient monitoring, and drug dispensing.

(c) Massive Machine-Type Communication (mMTC): The mMTC is designed to connect millions of IoT devices for long-range and low-power communication for continuous data exchange. It is often used for hospital automation, wearable health monitoring, home care monitoring, and hospital asset tracking.

3.1.2 Deployment Based Network Slicing

(a) Vertical Network Slicing: In virtual network slicing, the network (i.e., RAN, core) is portioned as per the requirement of use cases to address the Quality of Service (QoS) problem. In healthcare, virtual network slicing enables the hosting of customized slices for medical applications such as telemedicine, medical IoT, and AR/VR to facilitate real-time consultations, remote patient monitoring, real-time AR/VR rehabilitation programs, etc.

(b) Horizontal Network Slicing: The horizontal network slicing is responsible for handling network congestion at the edge of the network, ensuring computation offloading, and optimizing resource allocation by distributing computational tasks across multiple network layers. In healthcare, horizontal network slicing ensures edge-based AI diagnostics (i.e., CT scenes, MRI analysis), AR/VR-enabled remote surgery, and smart ambulances.

(c) Static Network Slicing: In static network slicing, slices are predefined, which means fixed, and are not adaptable to any changes as per network conditions. Operators must select the slices before hosting any application. In healthcare, static network slicing provides dedicated emergency responses and fixed bandwidth allocation.

(d) Dynamic Network Slicing: Based on the changing network condition, service demands, and resource availability, dynamic network slicing allows operators to dynamically design, customize, and deploy slices. In healthcare, dynamic network slicing allows remote ICU monitoring, Network Slice as a Service (NSaaS), and on-demand telemedicine.

3.2 Benefits for Healthcare Services

In healthcare, network slicing ensures reliable, secure, and low-latency connectivity for critical medical applications such as the emergency ambulance, remote patient monitoring, smart hospital management, and AR/VR medical training etc. (Trivisonno et al. 2017; Vergutz et al. 2020). Various application services with the benefit of network slicing in healthcare are explained as follows:

(a) Remote Patient Monitoring: The dedicated network slice allows the Internet of Medical Things (IoMT) devices to continuously monitor patients' health data, including blood pressure, heart rate, and glucose levels.
Requirement: URLLC.

(b) Emergency Ambulance Service: The dedicated network slice enables the development of a smart ambulance equipped with medical devices and communication tools to assist patients on a priority basis.
Requirement: URLLC and eMBB.

(c) Remote Consultation Service: The dedicated network slice can provide high-definition remote consultation to patients using video conferencing.
Requirement: mMTC.

(d) AR/VR Medical Training: The dedicated network slice can provide the necessary bandwidth for AR/VR application services (i.e., surgical procedures and medical training) to provide a hands-on learning experience.
Requirement: eMMB.

(e) Smart Hospital Management: The dedicated network slice allows IoT devices to monitor and manage various

services of the hospital, such as patient tracking, environmental control, and equipment management.
Requirement: eMBB, mMTC, and URLLC.

(g) Enhanced Data Privacy and Security: The dedicated network slice ensures an isolated network environment using specific security protocols for different services, reducing the risk of unauthorized access and data breaches.
Requirement: Dedicated security protocols, healthcare regulations, and standards.

3.2.1 Applications of Network Slicing in Heathcare

In Kapassa et al. (2019), the authors proposed an eHealth system that collects heterogeneous data from a variety of medical IoT devices powered by 5G network slices. In this, the 5G network slice is utilized to enhance the quality of the process, predictive analysis, and visualization of patients' health records. These health records contain indoor and outdoor healthcare data of registered patients collected through medical IoT devices. In the future, a deep investigation into real-time slicing adoption across all levels of the architecture will be done based on the user's quality of experience. In Njah et al. (2023), the authors proposed an Intent-based Networking architecture for Healthcare 4.0. Further, the natural language processing-based intelligent intent refinement is designed to extract and map keywords for the network policy model. In the future, optimization strategies such as pruning and parallelism will be implemented to enhance the speed of the proposed architecture without affecting the accuracy. In De Brito et al. (2021b), the authors presented a securing eHealth network for a Non-Public Network (NPN) hospital environment by applying network slicing and BC to ensure data integrity and reliability. In the future, proof-of-concept will be deployed to assess its feasibility. In Roddy et al. (2019), the authors proposed a PoC for 5G network slicing to mission-critical eHealth use case (i.e., ambulance telemedicine system) in the SliceNet project. Further, it demonstrates the collection, processing, and streaming of patient data using a 5G-connected health slice with edge computing to assist in assessing a potential patient suspected of having a stroke. In this, AI/ML image processing is involved to provide hospital clinicians and paramedics with in-ambulance diagnostics that enhance and improve patient treatment pathways. In Slimen et al. (2021), the authors proposed a cognitive-based system that collects and monitors vertical data to estimate the Quality-of-Perception (QoP) in a 5G slice for eHealth. Further, an ML model is presented to analyze the QoP received from vertical users to show the interaction between different components of the proposed system. In the future, a prediction model will be integrated to detect anomalies at E2E slices. In Mavrogiorgou et al. (2019), the authors presented a 5G network slicing-enabled end-to-end platform in which IoMT devices target three different types of medical applications, i.e., Massive IoMT (collect patient's data), enhanced mobile broadband (remote doctor examination), and mission-critical services. Initially, the data acquisition mechanism is implemented, which collects the data from IoMT and further provides it to the slicing management mechanism for segmenting the data and passing it to different network slices as per the computational requirement. In the future, the platform will be tested with additional heterogeneous IoMT devices and the implementation of visualization modules through a web interface. In De Brito Gonçalves et al. (2021a), the authors presented an NPN architecture in an eHealth environment based on 5G network slicing that ensures QoS and privacy requirements among network slices. Further, a public BC is incorporated for decentralized storage that receives data from Oracle to prevent an unauthorized privilege escalation. In the future, the integration of the IPSec and privacy module will be done. In Wang et al. (2019), the author proposed a SliceNet framework for media-centric eHealth (smart ambulance, emergency telemedicine) applications using 5G network to address QoS (bandwidth, delay/latency) and low latency requirements. In the future, integration of the connection algorithm will be done to improve QoE for eHealth applications.

4 Integration of Blockchain Technology and Federated Learning

The FL or Decentralized Learning (DCL) is a cutting-edge paradigm of AI-ML that encourages decentralized training models across multiple devices or institutions, parallel with localizing the data. This concept resolved the challenge of scaling storage infrastructure due to an exponential growth in the data. Another major issue addressed by FL is data privacy thus ensuring that sensitive data remains on the participant's device or within the institution. FL works on the architecture distributed ML architecture focuses mainly on distributed computing and data privacy protection that helps to prevent repetitive redundancy in data storage (Siniosoglou et al. 2023). Client-uploaded model updates without authentication pose significant challenges within the FL network despite its advantages in the AI domain. These challenges include establishing trust among participants, addressing security vulnerabilities, and facilitating efficient communication across distributed networks.

4.1 Challenges and Solutions

While FL offers numerous advantages, it also comes with significant shortcomings, which is a critical matter if data and models are both very sensitive. Major challenges in FL are listed as follows (Yaacoub et al. 2023; Yurdem et al. 2024; Wen et al. 2022):

(a) Concern in Data Security

- The data exposure risk within the distributed network or nodes exchanging model updates face potential unauthorized access to sensitive operational data. The aggregation point could potentially analyze shared information to extract confidential details, compromising data security.
- The malicious activity done by bad actors can disrupt the system by introducing corrupted training samples or manipulated models into the network. Additionally, dishonest participants may strategically delay operations or break commitments, compromising system reliability.

(b) Concern in Participation Framework

- The engagement barriers of the distributed learning system assume that all network nodes are willing to participate. However, this is an unrealistic expectation without proper compensation for their data and computational contributions. If the benefits are unclear, some nodes may choose to opt out entirely.
- The reward distribution is a challenge in FL due to distributed architecture. This makes fair compensation complex. Current frameworks lack standardized methods for reward allocation, leading to potential inequities. Additionally, these systems often fail to implement clear mechanisms for recognizing valuable contributions or addressing non-compliance, further complicating fair reward distribution.

(c) Concern in System Resilience and Performance

- The system depends heavily on a single aggregation node, making the central point vulnerable. If this node is compromised, the entire system gets disrupted.
- The network lacks robust protection measures without standardized security protocols. Malicious actors can tamper with model updates, and the system lacks effective tools to detect and remove these threats, which potentially degrade the overall performance.
- Many implementations using central coordination points lack clear audit trails and verification mechanisms for generated models, this overall affects the trustworthiness.
- The distributed architecture requires frequent data exchange between numerous participants and the coordination point. This high communication volume often strains network resources, particularly with bandwidth limitations.

4.1.1 Need of Blockchain in Federated Learning

BC and FL are complementary technologies that, when combined, address critical challenges in data security, privacy, and decentralized decision-making in modern technological ecosystems (Deng et al. 2022; Musa et al. 2023).

BC provides a revolutionary approach to many of the bottleneck challenges of AI/ML. As a decentralized and immutable record keeper, BC ensures secure and transparent record-keeping, which encourages trust among participants of FL ecosystems. By integrating BC with Federated Learning (FedBLOC), stakeholders can collaboratively train ML models without the need for the supervision of central authority, ensuring both privacy and accountability. BC is primarily a data structure architecture that ensures secure, decentralized, and immutable data storage. The key technologies in BC include smart contracts, consensus algorithms, hash functions, and digital signatures. These concepts enable data consistency, multi-party consensus, and redundancy in storage, thus making BC highly resistant to tampering and fraud (Shrimali and Patel 2021).

In a BC network, data is stored in a distributed manner, meaning no single participant has complete authority over stored information. Instead, each user of the network, also referred to as a node, validates Txs through a consensus mechanism before they are added to the BC network (Musa et al. 2023). This approach enhances transparency and promotes trustless authentication, where no central authority is required for verification. BC's decentralized nature ensures that all participants maintain a copy of the ledger, thereby eliminating single points of failure and enhancing security. Furthermore, BC promotes anonymity while maintaining transparency. Each exchange is recorded on a public ledger, together with the identities of the participants remain pseudonymous. The system relies on cryptographic techniques to guarantee that Txs remain secure and immune to tampering. Additionally, redundant data storage across multiple nodes prevents data loss and ensures continuous availability, making FedBLOC a robust solution for applications requiring trust, security, and decentralization.

4.1.2 Classification of FedBLOC

FedBLOC is a hybrid approach that integrates the benefits of public and private BC along with the advantages of FL. It is particularly useful in scenarios where a group of organizations or entities want to collaborate while maintaining control over their data and processes. FedBLOC can be classified based on various criteria, such as the consensus mechanism, access control, and governance model.

(a) Consensus Mechanism: The consensus mechanism refers to the process by which nodes (participants) in a BC network collaborate to validate Txs and agree on the state of the ledger. In a FedBLOC, consensus participation is limited to a pre-approved group of nodes, ensuring efficiency, control, and trust among participants. Controlled by preselected nodes rather than being fully open like public BCs. The nodes are classified as validators and observers. Validators can propose and validate Txs, while observers can read the ledger but cannot participate in consensus. The protocols used to achieve agreement are PBFT, PoA, RAFT, etc. This strategy ensures efficiency, control, and trust, making it ideal for consortium-based applications like supply chain management, healthcare, and finance.

(b) Access Control: Access control in FedBLOC classifies the actions of each participant in the BC network. Unlike public BC or private BC, FedBLOC operates on a consortium model, where a group of pre-approved entities controls the network. Permission levels ensure that only trusted participants can validate Txs, access data, or perform specific operations. Participants are divided into validator (full participation), observer (partial participation) and clients (restricted participation) to ensure security and efficiency.

(c) Governance Models: The governance model in FedBLOC determines how decisions are made, rules are enforced, and how participants interact within the network. The two primary governance models are Centralized and Decentralized. In a centralized governance model, a single entity or a small group of entities has control over the Federated BC network. This model is often used in consortium-based systems where trust is placed in a central authority. For example, a consortium of banks led by a central bank that controls the rules and operations of the BC network. In a decentralized governance model, decision-making power is distributed among all participants in the Federated BC network. This model promotes fairness, transparency, and inclusivity, for example a consortium of healthcare providers where each hospital, clinic, and insurer has an equal vote in network decisions.

4.1.3 Architecture Layers of FedBLOC

FedBLOC architectures integrate elements of FL and BC technology, creating secure, decentralized systems for data

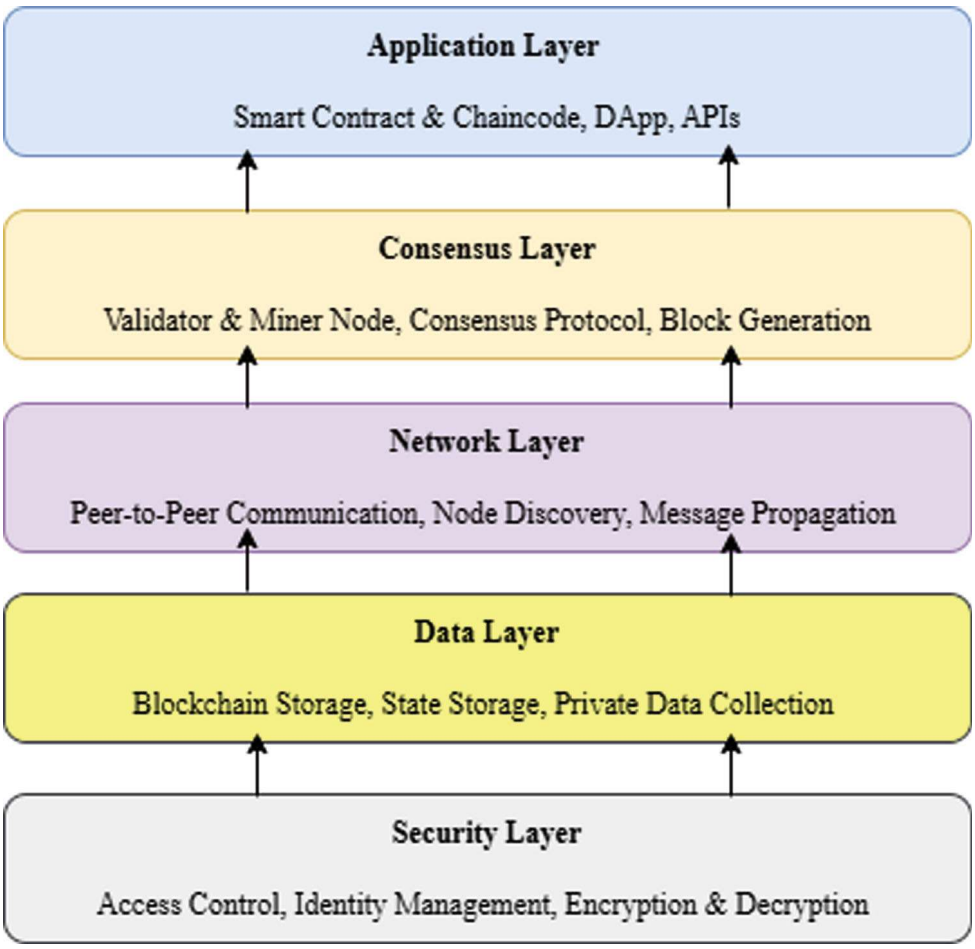


Fig. 6 Layer architecture of FedBLOC

sharing, model training, and consensus. The description of different layers and their functions associated with FedBLOC architecture is shown in Fig. 6. Each layer plays a critical role in ensuring the BC's functionality, scalability, and security, making Federated BCs suitable for consortium-based applications across industries (Ning et al. 2024; Calo and Lo 2023; Salim et al. 2024).

(a) Application Layer: The application layer forms the topmost layer of a federated BC architecture, serving as the interface between end-users and the underlying BC infrastructure. This layer handles smart contracts, DApps, and APIs for user interaction and business logic. Smart contracts represent self-executing code that implements custom business logic, manages access control rules, handles state transitions, and processes events within the BC network. These contracts automate complex business processes while ensuring transparency and immutability. Decentralized Applications (DApps) provide user-friendly interfaces that interact with these smart contracts, and business-specific applications designed for industries. It integrates with existing systems and utilizes client-side libraries for development. APIs facilitate interaction with BC through RESTful interfaces, GraphQL endpoints, SDK integration points, and event subscription services, enabling real-time notifications and seamless integration.
(b) Consensus Layer: The consensus layer manages the critical aspects of agreement and validation across the network and ensures agreement among nodes through protocols and block production. Validator nodes form the backbone of this layer, implementing sophisticated block validation logic, maintaining Tx order, verifying network state, and participating in leader election protocols. The consensus protocol, which might implement PBFT, Raft, or PoA, ensures agreement across all participating nodes while maintaining network security and efficiency. Block production mechanisms govern the creation and validation of new blocks, implementing rules for Tx batching, block finalization, and chain extension while ensuring the integrity of the BC.
(c) Network Layer: The network layer handles all aspects of inter-node communication and network maintenance. P2P Communication protocols manage direct node-to-node messaging, maintain network topology, handle connection pooling, and implement protocol-level security measures. Node discovery mechanisms combine static configuration with dynamic peer discovery, managing network membership and monitoring node health. Message propagation systems ensure efficient distribution of Txs and blocks across the network, implementing sophisticated gossip protocols for state synchronization and network-wide consistency.
(d) Data Layer: The data layer manages the persistent storage, state management, and private data handling of BC data. BC storage handles the fundamental data structures for blocks, Tx records, chain states, and historical data management. The state database maintains the current world state, implements key-value stores for efficient data access, supports rich queries for complex data retrieval, and manages data versioning. Private data collections enable secure management of confidential information, implementing sophisticated sharing rules, access patterns, and data segregation mechanisms to ensure privacy while maintaining network transparency where appropriate.
(e) Security Layer: The security layer provides access control, identity management, and encryption for secure operations and implements comprehensive protection mechanisms across the network. Access control systems manage permissions through Role-based Access Control (RBAC), implementing both organization-level and network-level policies to ensure appropriate resource access. Identity management systems utilize certificate authorities and PKI infrastructure to verify and manage participant identities while maintaining secure credential management. Encryption mechanisms protect data at rest and in transit through channel-level encryption, transport layer security, private data encryption, and sophisticated key management systems, ensuring the confidentiality and integrity of sensitive information across the network.

4.2 Use Cases and Benefits

This section presents real-world use cases of FedBLOC architectures, highlighting the problems solved, the architecture employed, and the benefits offered.

4.2.1 Healthcare: Secure Federated Learning for Medical AI

Training AI models to detect cancer through medical imaging by combining datasets from multiple hospitals to improve model accuracy (Eldin et al. 2023) using permissioned federated BC. The healthcare industry deals with highly sensitive patient data distributed across hospitals, clinics, and research institutions. Federated BC architecture enables privacy-preserving collaboration to build AI models for applications such as disease detection, treatment planning, and drug discovery.

(a) Hospitals act as nodes in an FL system, training models locally on their patient data.
(b) Model updates (not raw data) are securely shared with other nodes using a BC network.
(c) BC ensures tamper-proof logging of updates and tracks contributions from each institution.

(d) Smart contracts automate incentive mechanisms, rewarding nodes based on their contributions to the global model.

The integration of FL and BC technology has gained significant attention in healthcare, due to improving data privacy, security, and interoperability. The authors in Myrzashova et al. (2023), provide a comprehensive review to explore the convergence of BC and FL in smart healthcare systems. The main issue addressed is how FL facilitates decentralized ML without compromising patient data privacy. The research also examines BC's role, challenges, and opportunities in providing a secure framework for such collaborative models. Focusing on healthcare-based cyber-physical systems, Liu et al. (2023) discussed an incentive mechanism that is fair and based on points for contributing participants. The authors in Kim and Hong (2019), discussed the BAFL model using a hash algorithm, the research mainly focuses on securing FL in Industrial Internet of Things (IIoT) settings by addressing vulnerabilities such as model tampering, data falsification, and privacy leakage. The approach introduces a Chameleon Hash Scheme with a Variable Trap Gate (CHCT) to enhance security. In Jiang et al. (2022), the authors suggested a transmission mechanism to solve network resource allocation problems to achieve reliable two-sided digital architecture in the digital twin. With the growing adoption of BC-based FL in IIoT, IoV, and medical systems, FedBLOC architecture is certainly gaining scholarly recognition. Future research must focus on enhancing privacy, security, global and transmission mechanisms, and personalization to advance its real-time deployment and application.

4.2.2 Benefits of Federated Learning in Healthcare

(a) Data Privacy and Improved AI Model Performance: The FL ensures that each patient's data remains local so that the confidentiality of the patient data is preserved. Since FedBLOC provides an opportunity for each institution to train their AI models without compromising sensitive information, aligning with ethical and legal privacy concerns. Therefore, each model behaves as an independent identity and can improve performance without any reliability on center datasets. This is particularly more beneficial in medical applications where variability in patient demographics and conditions is crucial for reliable predictions.
(b) Security: In the healthcare industry, the security of data and the tampering of the model is a primary concern, and the BC's immutability prevents unauthorized modifications, making the system more resistant to cyber threats and fraud. This added layer of security builds trust among healthcare organizations and patients.
(c) Collaboration: Collaboration is crucial in healthcare, particularly for studying rare diseases where patient data is scarce and scattered across multiple institutions. Traditional AI model training requires pooling patient data in a central repository, which raises privacy concerns, regulatory challenges, and logistical difficulties. FL combined with BC offers a solution by enabling institutions to collaboratively build AI models without sharing raw data while ensuring security, transparency, and regulatory compliance.

5 Summary

This chapter investigates the integration of BC and FL to enhance the management of network slices in 5G, Edge, and Cloud environments, particularly for healthcare applications. It begins with an overview of BC, detailing its types (public, private, consortium, hybrid), components (smart contracts, consensus protocols), and healthcare use cases, such as secure EHR management and tamper-proof data sharing. FL is introduced as a decentralized ML approach that preserves data privacy by training models locally on Edge devices. The paper classifies FL algorithms as weighted, adaptive, momentum-based, secure, and quantized—and highlights their applications in medical AI, such as cancer detection and pandemic response.

Network slicing, a key 5G feature, is analyzed for its ability to create virtualized, QoS-aware networks tailored to healthcare needs. Service-based slices (eMBB, URLLC, mMTC) and deployment-based slices (vertical, horizontal, static, dynamic) enable applications like remote surgery and IoT-driven hospital automation. The integration of BC and FL into FedBLOC architecture addresses challenges such as data security, trust deficits, and communication inefficiencies. FedBLOC utilizes BC's immutability for auditability characteristic with localized training for privacy, enabling collaborative AI model development across hospitals without raw data exchange. Use cases include secure model updates, automated incentive distribution via smart contracts, and tamper-resistant logging of contributions.

Despite its promise, challenges like Byzantine attacks, reward inequity, and scalability require further research. The chapter concludes by emphasizing the transformative potential of combining BC, FL, and network slicing to build resilient, privacy-centric healthcare systems, urging future work on adaptive consensus mechanisms and cross-domain interoperability.

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